



Exploiting stance similarity and graph neural networks for fake news detection

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ABSTRACT

The widespread dissemination of fake news on social media has substantial economic and social implications. Although traditional propagation-based methods employing graph neural networks show promise for fake news detection, they disregard the influence of confirmation bias in the spread of fake news between users with similar viewpoints. This paper presents a detection approach that accounts for opinion similarity between users by scrutinizing their stances toward news articles and users' post interactions. Using a graph transformer network, our method simultaneously extracts global structural information and interactions of similar stances. Furthermore, it addresses the challenges of stance analysis targeting microblogs while minimizing the effect of poorly represented stance features. We evaluated our approach using custom-crawled Twitter data and the benchmark FibVID dataset. It demonstrated a marked improvement in detection performance compared with conventional methods, including state-of-the-art methods.

1. Introduction

The rise of social media platforms, such as Twitter and Tencent Weibo, has facilitated the rapid dissemination and collection of information. However, the spread of fake news through these platforms has misled readers and negatively affected society at large. As such, there is a pressing need to counteract the propagation of fake news on social media. Nonetheless, human effort alone cannot effectively control the spread of fake news, which has been reported to propagate more rapidly than true information. This situation underscores the necessity of developing an efficient fake news detection system.

Fake news detection methods are generally classified into two major categories: content-based and context-based methods. Content-based detection methods use sentences and other elements of news articles to identify and comprehend semantic relationships [1]. However, a common challenge to content-based methods is that fake news often mimics real news, complicating detection based solely on content [1]. Consequently, it is essential to consider other features derived from the social aspects of news articles [2]. In contrast to content-based detection methods, context-based methods [3] can identify news independently of its content by analyzing social media interactions. This paper focuses on a specific context-based method known as the propagation-based method. Our approach leverages the distinct propagation patterns of fake and real news, extracting features from the diffusion structure for detection [4]. Most modern propagation-based methods use a graph

representation in which nodes represent news articles and users, and edges represent news sharing. These methods extract features from the graph structure using graph neural networks (GNNs).

No previous propagation-based detection methods have solved the problem of misidentification caused by intricate propagation structures. Specifically, large-scale complex networks and graphs with similar propagation structures for real and fake news are generated by users and bots manipulating the structure of the news propagation graph (e.g., by posting a large volume of spam), leading to misidentification [5].

An effective solution to this problem is the detailed exploration of structure formation processes and the enhancement of structure identification capabilities. Previous studies have shown that fake news spreads through a chain of shares between users who follow similar principles [6]. This propagation pattern can be traced to confirmation bias, whereby users are inclined to believe information that aligns with their own principles [7]. According to these findings, we can assume that structure identification performance may be improved by considering stance similarity in news-sharing interactions. However, traditional propagation-based methods overlook this phenomenon and focus solely on structural differences.

We propose a novel propagation-based fake news detection method capable of extracting stance similarity during news sharing. Specifically, we fine-tune the bidirectional encoder representations from

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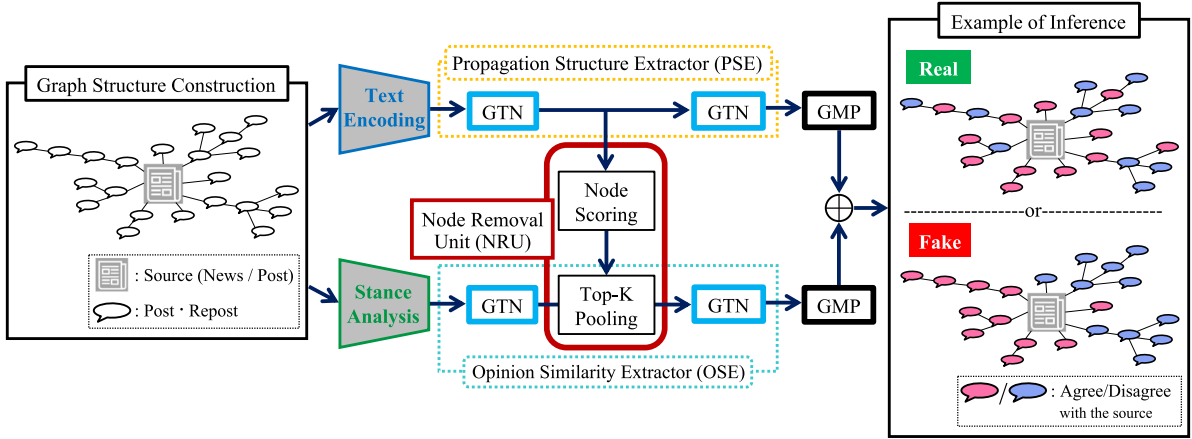


Fig. 1. Model overview and inference example (number of layers $L = 2$). The text of each node is encoded and analyzed using BERT and stanceBERT, which is BERT fine-tuned for the stance analysis task. The example illustrates that fake news tends to be shared between users who have similar stances.

transformers (BERT) natural language processing model [8] for stance analysis tasks, extract user stances as embedded representations, and use them as feature vectors of nodes. Moreover, we employ the graph transformer network (GTN) [9] to calculate the similarity between the feature vector of a specific node and the features of its neighboring nodes. In this manner, we can extract the user stance as an embedded representation, thus allowing the GTN to extract overall structural information and similar stance interactions simultaneously. This method enables the detection of similar stance propagation due to confirmation bias, which is a characteristic of fake news propagation. To address the challenges to stance extraction in microblogs such as Twitter, that is, the lack of text for feature representation learning, our method also facilitates the removal of nodes with characteristics that obstruct the extraction of stance similarity.

The contributions of this paper are as follows:

1. We enhance the learning of fake and real news propagation structures through multi-aspect learning, which incorporates text and stance encodings during news sharing. Our method boosts the performance of structure representation learning using GNNs by eliminating nodes whose stance representation is inadequate.
2. We describe a unique dataset of news propagation on Twitter that is based on the most recent fact-checking data. We compare its performance with a benchmark dataset, the FibVID dataset [10], using performance comparison experiments. The results indicate that our proposed method demonstrated superior discriminatory performance on real-world data, thereby confirming its effectiveness for real-world fake news detection.
3. We provide a detailed description of the fine-tuning process for BERT and the GTN approach on the stance analysis task. Furthermore, we validate the effectiveness of our proposed method through the results of experiments and comparisons with conventional methods, including state-of-the-art methods. We believe that this paper offers valuable insights and directions for future research on fake news detection.

2. Related work

2.1. Content-based detection

Content-based methods are detection strategies that use various article elements, such as text, titles, and images [3]. They can be employed to verify the authenticity of news content and to detect fake news by identifying emotionally charged language [11]. The primary focus of these methods lies in the article text, which is analyzed

to discern the unique writing style often associated with fake news. Recent studies have incorporated natural language processing models to extract semantic content from the article text, with some models capturing semantic relationships among multiple article elements. Zhou et al. [12] suggested a detection model based on convolutional neural networks that uses the article text and associated images. Palani et al. [13] obtained richer semantic features from the text of articles by using BERT [8], which outperformed the sentence generation accuracy of previous sequential processing models.

2.2. Context-based detection

Context-based detection employs social media interactions to discern anomalies [3]. Key social media interactions encompass following-follower relationships and news propagation, indicating the sequential dissemination of news or posts.

Attributes such as follower counts and user profiles can help capture the characteristics of users who are actively involved in spreading fake news. Lu et al. [14] constructed a friendship network, representing each user as a node and connecting it to those in following-follower relationships by edges. Friendship relationships were deduced by interchanging user attributes as node features using GNNs. Similarly, user attributes have found application in non-GNN methods [15,16]. However, the arbitrariness of user attribute selection across methods defies standardization. Consequently, they are impractical because of the extensive time and labor required for collecting them from social media, particularly when using attributes whose contribution to detection is uncertain, such as location, or those that are inaccessible on specific social media platforms, such as age.

We propose the extraction of the stance of news and posts as an alternative method of capturing user characteristics. The stance refers to the intensity of opinions that users express on a news topic, such as agreement or disagreement [3,17]. In a pioneering attempt to incorporate stance analysis into fake news detection, Nguyen et al. [18] introduced a learnable stance loss, centering on the commonality of stance between the news and associated posts.

2.3. Propagation-based detection with GNNs

The news propagation-based method detects fake news by constructing a graph composed of nodes representing news and posts, and edges representing the sharing of news and posts. This method applies the observation that fake news propagation involves more users and forms a deeper structure than real news because of its cascaded spread from one user to another [4]. Early research on propagation-based methods proposed using a time-series processing model to extract propagation

information [19]. However, this approach struggled to identify structural information because propagation was represented as time-series data, in which news and posts were arranged chronologically.

Since the emergence of GNNs, detection models integrating structural discernment have become popular. Such detection models work as follows: (1) Text features, extracted from articles and posts for news detection, are initialized as the feature vectors of the corresponding nodes. (2) The news propagation graph is then fed into the GNN, which aggregates and updates node features according to the structure of the graph. (3) Using global mean pooling (GMP), features from all nodes are averaged, yielding the entire graph's structural information as a feature vector. (4) These structural features are then passed through a multilayer perceptron (MLP) to distinguish between real and fake news.

Monti et al. [20], in the first study to apply GNNs to propagation-based methods, used the GNN baseline, the graph convolutional network (GCN) [21], and introduced a detection model that follows the aforementioned four steps. Dou et al. [22] enhanced detection accuracy by averaging the textual features of users' most recent posts to extract user preferences and combining propagation structure information with a textual representation of the article body. Bian et al. [23], in their pioneering work applying GNNs to rumor detection, implemented top-down and bottom-up aggregations to emphasize source tweet features. Wei et al. [24] increased accuracy by introducing unsupervised edge loss to optimize edge weights. Similarly, Tian et al. [25] constructed a network consisting of posts and their replies, using BERT in conjunction with a graph attention network (GAT) [26] to discern subtle relationships between opinions expressed in parent and child reply posts.

3. Proposed method

Fig. 1 presents an overview of the proposed architecture, which comprises the following components: propagation graph construction, a propagation structure extractor (PSE), an opinion similarity extractor (OSE), a node removal unit (NRU), and propagation graph classification.

3.1. Preliminaries

3.1.1. Graph construction

We construct two identical undirected graphs, which we explain here using Twitter as a representative example. We designate a node representing a news item as the root node. Each news-sharing tweet creates an edge between the tweet node and the news node. Subsequent retweets of this tweet lead to the creation of additional edges between the retweet nodes and this tweet node, ultimately forming a news propagation graph. The body text of the article is input into BERT to obtain a text representation $\hat{\mathbf{t}}_0$, which serves as the root node's feature. Likewise, tweet and retweet body texts are input into BERT, with the outputs assigned to the corresponding nodes. This process yields the news propagation graph $\hat{G} = (V, E, \hat{X})$, where V denotes the set of node indices, E denotes the set of edges, and $\hat{X} = [\hat{\mathbf{t}}_0, \hat{\mathbf{t}}_1, \dots, \hat{\mathbf{t}}_N]$ denotes a feature matrix composed of node feature vectors, with N representing the total user count.

We apply stance analysis to news and users to extract interactions among users who share similar opinions. This involves fine-tuning BERT using a stance analysis dataset; the fine-tuned BERT is referred to as stanceBERT. Using stanceBERT, we perform stance analysis by generating a feature vector that represents the stance of the input text. Details of the fine-tuning process, including the stance analysis dataset, are outlined in Section 4.3. By inputting news articles and tweets into stanceBERT and assigning its output to the corresponding nodes, we obtain the graph $\bar{G} = (V, E, \bar{X})$, where $\bar{X} = [\bar{\mathbf{t}}_0, \bar{\mathbf{t}}_1, \dots, \bar{\mathbf{t}}_N]$ denotes a feature matrix that represents the stance characteristics of the news articles and tweets.

3.1.2. Graph transformer networks

GNNs learn graph structures by repeatedly aggregating features of neighboring nodes. Specifically, a GNN layer has an aggregation function, which gathers features of the neighbors of each node, and a combination function, which updates each node's features to features that reflect the graph structure by combining the aggregated features for each corresponding node. By stacking multiple layers and iteratively performing neighborhood aggregation, it becomes possible to access information of nodes that are distant from the node of interest, thereby enabling extraction of the global structure of the graph.

Various GNN models with different aggregation methods have been proposed since the advent of GCNs. The GTN is the first GNN model to apply a Transformer to neighborhood aggregation. It calculates the similarity between each node's features and its neighboring nodes' features using attention, and aggregates the features of neighboring nodes weighted by their similarity.

For a GTN model with L layers, the aggregation and combination for a target node i ($= 0, 1, \dots, N$) in the l th layer of the GTN is formally expressed as follows:

$$\mathbf{m}_i^{(l)} = \text{aggr}(\mathbf{h}_j^{(l-1)}; j \in \mathcal{N}_i) = \sum_{j \in \mathcal{N}_i} \alpha_{ij}^{(l)} W_1^{(l)} \mathbf{h}_j^{(l-1)}, \quad (1)$$

$$\mathbf{h}_i^{(l)} = \text{comb}(\mathbf{h}_i^{(l-1)}, \mathbf{m}_i^{(l)}) = \sigma(W_2^{(l)} \mathbf{h}_i^{(l-1)} + \mathbf{m}_i^{(l)}), \quad (2)$$

where $\mathcal{N}_i$ denotes the set of indices of neighbors of the node i . $\mathbf{h}_i^{(l)}$ denotes the node feature vector that node i possesses, which corresponds to the output of the l th layer of the GTN. $W_1^{(l)}$ and $W_2^{(l)}$ are learnable weights, and $\sigma(\cdot)$ is the ReLU activation function. $\alpha_{ij}^{(l)}$ is an attention coefficient calculated from the similarity between the features of the node of interest and the features of neighboring nodes, and is determined as follows:

$$\alpha_{ij}^{(l)} = \frac{\langle W_3^{(l)} \mathbf{h}_i^{(l-1)}, W_4^{(l)} \mathbf{h}_j^{(l-1)} \rangle}{\sqrt{d}}, \quad (3)$$

$$\alpha_{ij}^{(l)} = \text{softmax}(\alpha_{ij}^{(l)}) = \frac{\exp(\alpha_{ij}^{(l)})}{\sum_{j \in \mathcal{N}_i} \exp(\alpha_{ij}^{(l)})}, \quad (4)$$

where $\langle \cdot, \cdot \rangle$ denotes the dot product. $W_3^{(l)}$ and $W_4^{(l)}$ are learnable weights. d denotes the feature dimension of $\mathbf{h}_i^{(l-1)}$. Hereafter, we represent Eq. (2), describing the GTN operation for the l th layer node feature matrix $H^{(l)} = [\mathbf{h}_0^{(l)}, \mathbf{h}_1^{(l)}, \dots, \mathbf{h}_N^{(l)}]$, as $H^{(l)} = \text{GTN}(V, E, H^{(l-1)})$.

3.2. Opinion similarity extractor

Interactions between users who share similar opinions can be characterized using a GTN. The GTN calculates the similarity between the stance features of a focal node and those of its neighboring nodes by following the structure. Formally, if $\bar{H}^{(0)}$ is the input to the OSE and $\bar{X}$ represents the stance features of graph $\bar{G}$, then we can say $\bar{H}^{(1)} = \text{GTN}(V, E, \bar{X})$. This mechanism enables us to consider the behavior of individuals who propagate fake news by believing and sharing opinions that align with their own beliefs; this is a phenomenon attributed to confirmation bias. Furthermore, the mechanism also allows for the extraction of complex structural information.

3.3. Propagation structure extractor

A GTN is also used to extract global features regarding propagation structures. The Transformer architecture within the GTN is intended to alleviate over-smoothing [27], thereby facilitating the extraction of effective global features [28,29]. For this reason, GTNs are adopted to extract a greater amount of global structural information than that extracted by traditional propagation-based methods. Inside the PSE, the graph $\bar{G}$ is provided as input, and the node feature matrix at the l th layer, $\hat{H}^{(l)} = \text{GTN}(V, E, \hat{H}^{(l-1)})$, is obtained. $\hat{H}^{(0)} = \hat{X}$ is the input to the first layer. It should be noted that $\hat{H}^{(1)}$ is then supplied as input to the subsequent NRU.

3.4. Node removal unit

When working with microblogs, such as Twitter, stance analysis is inevitably challenging. This partly stems from the difficulty of extracting sufficient features for stance identification from concise posts such as tweets. Additionally, the insufficient extraction of stance features may impede the successful extraction of stance similarity with adjacent users in news-sharing relationships, potentially obstructing the extraction of propagation structures and adversely affecting detection. A direct solution could involve preprocessing to eliminate nodes with deficient stance analysis. However, this may result in disrupted edges and significantly affected propagation structures. Hence, it is necessary to remove such nodes during the learning process to reduce their impact.

The proposed NRU can identify and eliminate nodes with stance similarities that adversely affect detection during the learning process. In this mechanism, each node feature is converted to a score using learnable weights, and nodes with low scores are removed as they are found to adversely affect detection. This conversion and removal adopts the Top- K pooling method [30]. To focus on nodes with stance similarities that could have a detrimental effect, the NRU is placed immediately after the first layer ($L = 1$). This mitigates the negative influence of structure extraction being impeded at the superficial stage of the OSE network and facilitates more efficient learning.

To improve the extraction of propagation structures, node features $\hat{H}^{(1)}$, carrying propagation structure information, are converted to scores $s^{(1)} = [s_0^{(1)}, s_1^{(1)}, \dots, s_N^{(1)}]$ according to the learnable weight w_{scr} and the activation function $\tanh(\cdot)$. In the updated graph $\tilde{G} = (V, E, \tilde{H}^{(1)})$, the nodes corresponding to the top $K\%$ of scores are pooled using $s_i^{(1)} = \tanh(w_{scr} \hat{h}_i^{(1)})$. This operation is extracted by the function $TopK\ Pooling(\cdot)$. The graph elements after pooling are formally represented as follows:

$$V', E', \tilde{H}'^{(1)} = TopK\ Pooling(s^{(1)}, V, E, \tilde{H}^{(1)}), \quad (5)$$

where V' denotes a set of node indices corresponding to the top $K\%$ of scores, E' denotes a set of edges formed between the nodes included in V' , and $\tilde{H}'^{(1)}$ denotes a feature matrix composed of feature vectors associated with nodes belonging to the set V' . The graph $\tilde{G}' = (V', E', \tilde{H}'^{(1)})$ after node removal is input to the second layer of the GTN within the OSE, and is aggregated and updated along the new edges E' . This can be formally represented as $\tilde{H}''^{(l')} = GTN(V', E', \tilde{H}'^{(l'-1)})$, where $l' = 2, 3, \dots, L$.

During the learning process of our proposed model, nodes with stance similarity that hinder propagation structure extraction are removed using node features that have propagation structure information, allowing complex propagation structure information to be extracted without corrupting it.

3.5. Propagation graph classification

Regarding the output $\hat{H}^{(L)}$ of the final layer of the GTN in the PSE and the output $\tilde{H}''^{(L)}$ of the final layer of the GTN in the OSE, we derive feature vectors that represent the entire graph through GMP. Subsequently, these two feature vectors are concatenated to integrate the propagation structure information with the information on the interaction between like-minded users, and perform binary classification as follows:

$$\mathbf{p} = W_{fc} \left(\frac{1}{|V|} \sum_{i \in V} \hat{h}_i^{(L)} \oplus \frac{1}{|V'|} \sum_{i' \in V'} \tilde{h}_{i'}''^{(L)} \right), \quad (6)$$

where W_{fc} denotes the learnable weight of the fully connected layer. Binary classification is performed using the vector $\mathbf{p}$, which consists of prediction probabilities for fake news and real news. We can capture complex propagation structures by combining propagation structure information with information on the interaction between like-minded users for classification.

Table 1

Statistics of the graphs constructed from the two datasets.

Statistics	Dataset	
	Custom	FibVid
Average #num nodes	4770	172
Average #num edges	14 308	513
Average of max depth	2.357	1.834
Maximum value of max depth	5	6
Average node degree	5.914	5.823

4. Experiments

4.1. Dataset

The FakeNewsNet dataset [31], traditionally used for comparing fake news detection methods, now suffers considerable data losses due to the deletion of tweets and account privatization. Consequently, we removed the missing data and supplemented it by collecting Twitter data via Twitter API v2, adhering to the protocols used in the creation of the FakeNewsNet dataset. We focused on news that had been verified by the fact-checking site PolitiFact. We concentrated on news published between December 2020 and July 2022, gathering a month's worth of propagation by iteratively searching for tweets containing words that are frequently used in the articles and their quote retweets. Following [4], we merged the collected data {Real:57, Fake:105} with the FakeNewsNet PolitiFact graphs {Real:51}, from which retweets that are not quotes were excluded, to create the PolitiFact dataset {Real:108, Fake:105}.

In addition to FakeNewsNet, we employed the FibVID dataset [10]. This dataset is supplied to evaluate propagation-based methods, grounded on fact-checks of tweets by PolitiFact and Snopes on topics such as COVID-19 and politics. The relevance of the content to the fact-checked tweets is determined by the cosine similarity of the text vector computed by Word2Vec, according to which the tweets and their quote retweets are collected and the tweet node is defined as the root node, which represents the information source. However, this dataset includes tweets with minor similarities, prompting us to eliminate tweets (both real and fake) whose similarity is less than 0.3. We then balanced the data by reducing the number of fake tweets to equal the number of real tweets while maintaining the distribution of fake tweets across eight similarity ranges (0.3 to < 0.4, ..., 0.9 to < 1.0, 1.0). Finally, we excluded graphs from which it was impossible to construct a structure, such as those consisting only of root nodes, or those unable to encode source tweets, and used {Real:276, Fake:277}. The inclusion of propagation originating from tweets with low similarity is probably a consequence of intentional data selection performed with the aim of constructing a clean dataset, for example, by eliminating duplicated tweet content within propagation.

The graph statistics for each dataset are presented in Table 1.

4.2. Model comparison

4.2.1. Comparison methods

We conducted comparisons using content-based methods, such as BERT+MLP and Longformer+MLP, following [22,32], to highlight the advantages of propagation-based approaches. Subsequently, we compared our method against the following propagation-based approaches presented in [20,22–25]:

- (i) GNNFN (GNN Fake News; Monti et al.) [20]: A GNN-based baseline solely consisting of L -layer GCNs that focuses on propagation structure extraction.
- (ii) BiGCN (Bi-directional GCN; Bian et al.) [23]: A model featuring a structural extraction mechanism composed of L -layer GCNs for top-down extraction (from news sources to tweet users) and a parallel bottom-up extraction (from tweet users to the source).

- (iii) UPFD (User Preference-aware Fake News Detection; Dou et al.) [22]: This approach appends the embedding of the article body to the output of the structure extraction mechanism, which consists of L -layer GCNs identical to GNNFN. It assumes the article body reflects the credibility of the news piece.
- (iv) EBGCN (Edge-enhanced Bayesian GCN; Wei et al.) [24]: This model introduces a framework between the GCN layers of a BiGCN that refines the edge weights associated with nodes. It thereby regulates the influence of nodes that might obstruct detection.
- (v) BERT+GAT (Tian et al.) [25]: This strategy is designed to extract semantic relation information across posts for structural extraction. The two parent-child posts are input together into BERT, and then the graph to which the text embeddings of BERT outputs are assigned is input into L -layer GATs, which enables the extraction of semantic relations through the hierarchical application of attention mechanisms.

For [20,22–24], graph $\hat{G}$ is input into the GNN. For [22], because of the inability to glean user preferences owing to the substantial time that has elapsed since the original submission, we resorted to using the BERT embedding of the post text. For [25], we ensured a fair comparison by considering only the BERT+GAT module, used for the network structure extraction. Our method is referred to as “Proposed” in the tables.

In experiments involving the FibVID dataset, which does not include the articles, we excluded comparisons with content-based methods or [22] to ensure fairness. The study reported in [24] using FibVID incorporated supervised loss derived from the output of the first layer of the GCN and unsupervised edge loss for parameter updates.

4.2.2. Comparison methods for the ablation study

In these experiments, GNNFN [20] was used as the baseline for the ablation studies to validate the efficacy of our proposed method. In the ablation study, we compared our method with the following methods:

- (a) -NRU: The Proposed model with the NRU module removed.
- (b) -OSE-NRU: The Proposed model with both the NRU and OSE modules excluded.
- (c) GNNFN+ST: An -NRU model in which the GTN has been replaced by a GCN.

4.3. Experimental settings

Although a multilayered GNN is necessary for capturing the entire graph structure, the number of layers must be set properly to avoid feature averaging, which increases with network depth [27]. Table 1 shows that the custom data required two layers ($L = 2$) to capture the two-hop news-tweet-retweet structure and the FibVID data required one layer ($L = 1$) to capture the one-hop tweet-retweet structure [33]. The node retention rate, K , was 90% for our custom data and 98% for the FibVID data, as a consequence of its cleaner nature and because its nodes are smaller than those of our custom data. Both datasets had a seed value of 777. For our custom data, we set the batch size, learning rate, and number of epochs to 64, 0.002, and 60, respectively. For the FibVID data, these values were 120, 0.0002, and 120, respectively.

We implemented all GNN models using PyTorch Geometric. For our experiments based on [24], we used two RTX A6000 GPUs, whereas for all other experiments, we used two Tesla V100 GPUs. We employed the bert_base_uncased model of BERT and its fine-tuned variant. For the fine-tuning process, we used the FakeNewsChallenge-1 dataset¹ and the Debatepedia dataset [34] as the stance analysis datasets. This balanced the data skew and allowed us to acquire a fine-tuned BERT model with a dataset comprising 15,000 sentences in each stance category label:

Table 2

Model comparison and ablation study: accuracy and F1-Macro.

Model	Dataset			
	Custom	FibVid		
			Acc	F1-Ma
BERT+MLP	0.7322	0.7311	–	–
Longformer+MLP	0.6384	0.6360	–	–
GNNFN [20]	0.7183	0.7084	0.6604	0.6501
BiGCN [23]	0.7328	0.7214	0.6713	0.6708
UPFD [22]	0.7793	0.7781	–	–
EBGCN [24]	0.8031	0.7946	0.6315	0.6299
BERT+GAT [25]	0.7657	0.7640	0.7291	0.7286
Proposed	0.8641	0.8636	0.7980	0.7956

agreement, disagreement, and unrelated. By inputting the text from the acquired dataset into bert_base_uncased and fine-tuning it to predict the corresponding labels, we successfully generated stanceBERT.

For comparison, we calculated accuracy (Acc) and F1-Macro (F1-Ma) using fivefold cross-validation.

4.4. Results and discussion

4.4.1. Results

The results are presented in Table 2, and the best performance in each column is highlighted in bold. Our proposed method outperformed all conventional methods on both the PolitiFact and FibVID datasets. Specifically, it outperformed the highest-scoring traditional method by 6% on both datasets. The calculation of stance similarity, which is absent from traditional methods, provides crucial insight for classification. On our dataset, we compared the average cosine similarity between the stance features of neighboring nodes for both real and fake news. The results reveal that the average stance similarity for fake news was higher than that for real news (U-test results, $p < 10^{-5}$). This finding confirms the phenomenon whereby users with similar opinions propagate fake news.

4.4.2. Ablation study

First, in Fig. 2(a), the performance comparison between -NRU-OSE and GNNFN shows an enhancement in structural extraction due to the implementation of GTN, resulting in an improvement of about 4% with respect to both metrics. A comparison of the -NRU-OSE and -NRU results reveals that the introduction of the OSE module boosted performance by nearly 8% with respect to both metrics, roughly doubling the enhancement enabled by the GTN. Conversely, the comparison between GNNFN and GNNFN+ST demonstrates that merely incorporating stance features yielded an improvement of approximately 2%. This underscores the significance of stance similarity in structure identification. Furthermore, contrasting Proposed with -NRU shows that employing the NRU contributed to an additional 3% performance increase. The contribution of the NRU was further explored in the cross-validation rounds, where the accuracy improvement was most evident. For the graphs used for validation, excluding those consisting only of news and tweet nodes, we calculated the ratio of the number of tweets and retweets removed to the total number of tweets and retweets. The results show that the percentage of tweets removed was smaller for real news, whereas the percentage of retweets removed was smaller for fake news (t-test results, $p < 0.05$). This finding aligns with the pattern reported in [4], and suggests that the learning approach of preserving deep subtrees for fake news and pruning them for real news contributed to the increase in accuracy. Next, Fig. 2(b) shows that the stance similarity extraction considerably boosted detection performance, even though the graph structure of FibVID is less intricate than that of our custom data.

¹ <http://www.fakenewschallenge.org/>.

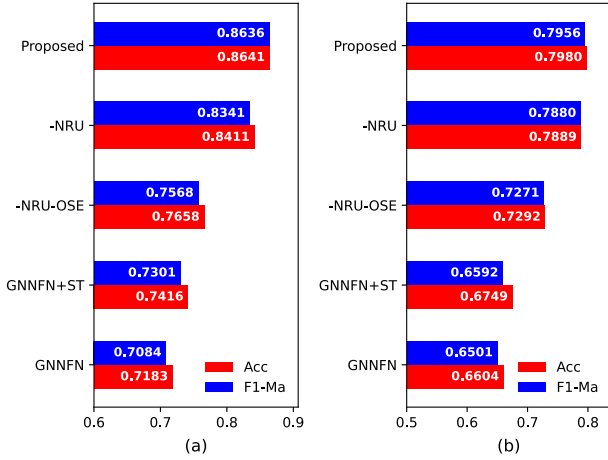


Fig. 2. Ablation results for (a) custom data and (b) FibVID.

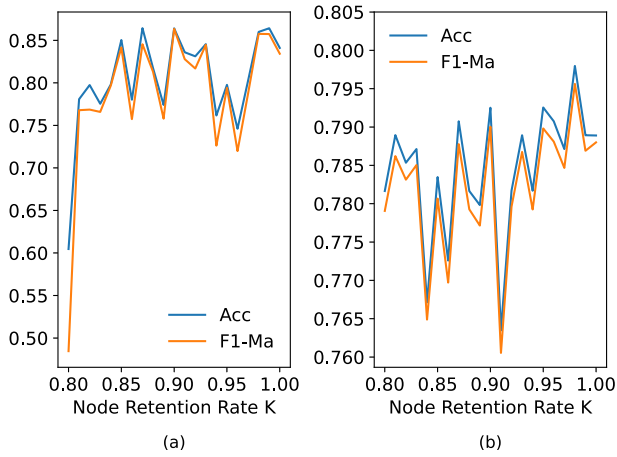


Fig. 3. Sensitivity of K for (a) custom data and (b) FibVID.

4.4.3. Sensitivity to K

Fig. 3 shows variations in both metrics as K ranges from 0.8 to 1.0 in 0.01 increments, with $K = 100\%$ corresponding to -NRU. Fig. 3(a) reveals substantial performance fluctuations. At $K = 80\%$, only nodes within a single hop were retained, and the second aggregation layer caused excessive smoothing, which caused the significant performance degradation. This indicates that feature propagation for a set number of layers is necessary to improve the performance of representation learning using GNNs, and a higher value of K is preferable. Performance matched or exceeded -NRU within a K range of 90% to 93%, reflecting a significant post-removal structural difference between real and fake news. Outside this range, the subtrees were either wholly removed or preserved, causing the difference in post-removal structure to be lost and leading to unpredictable training performance. A comparison between FibVID and our custom data highlights the reason that the fluctuation shown in Fig. 3(b) was less than that shown in Fig. 3(a): fewer nodes resulted in less change in the structure by node removal, and a larger amount of data resulted in sufficient learning of small changes in the structure. The K analysis suggests the problem of the NRU, in which the insufficiency of large-scale graph data makes it more difficult to distinguish post-removal structures, or mixing of

pre- and post-removal structural information in the OSE hinders NRU optimization.

4.4.4. Discussion and future work

The superior performance of BERT+GAT compared with -NRU-OSE indicates that the hierarchical extraction of parent-child topics captures relationships, such as stance. However, it is crucial to efficiently use the stance features in detection, as demonstrated by -NRU, which considerably enhances detection. Our experiments showed that amplifying homophily and heterophily is essential for improving propagation-based methods. The role of homophily in fake news propagation is recognized [35], whereas heterophily is associated with the reduction of over-smoothing [29]. Because stance features are homophilous elements projected into a feature space expressing a particular stance, they remain susceptible to feature smoothing. By adopting a GTN to reinforce heterophily, we achieved significant improvements in detection.

Of the source-content-aware methods, that of [22] tends to lean toward content-based detection and further performance improvements cannot be expected. The method of [23] seems to lean toward propagation-based detection. Although [23], which can adequately consider content-based features, holds promise, performance improvement is not expected on datasets such as FibVID, in which tweets are the source information, because of the complexity of extracting text features from microblogs.

We identify strategies for enhancing learning performance by comparing [20,24], and our proposed method. Specifically, we propose the reduction of the influence of nodes that impede learning and subsequently facilitate the GNN to learn the post-reduction state. Moreover, we suggest introducing an optimization process specifically for influence reduction, incorporated into the loss term. To solve the NRU problem, we will introduce the loss for NRU optimization separately from classification loss in the future. This introduction may make it possible to optimize the NRU by learning the post-removal structure and this learning may enrich the structural data for training.

5. Conclusions

This paper proposed a propagation-based fake news detection method that extracts stance similarity and news-sharing data. By examining the stance of each user during news dissemination and using a GTN to measure stance similarity between users, our method targets the spread of similar stances, a common trait in fake news propagation due to confirmation bias. This approach enables our method to navigate the intricacies of propagation structures and outperform traditional methods.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Soh Yoshida reports a relationship with Japan Science and Technology Agency that includes: funding grants. Soh Yoshida reports a relationship with Japan Society for the Promotion of Science that includes: funding grants.

Data availability

The authors do not have permission to share data.

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